

Internet Appendix for

“Belief Records and the Price of Disaster Fear”

Appendix IA.1. The physical-minus-priced tail gap

The model does not deliver the procyclical *sign* of the physical tail from its primitives. The realized physical tail carries the belief-revision jumps, whose cyclicity is a tradeoff the fixed-law economy leaves open: revisions cluster at disasters (countercyclical in frequency), but complacency makes them largest out of long calm spells, where the held views sit farthest from the truth (procyclical in size); whether size dominates frequency requires tying the cycle to the incumbent paradigm, an equilibrium step beyond the economy priced here. What the model *does* give is a disciplining path. Read against the directly priced, countercyclical intensity, the procyclical physical tail of Gormsen and Jensen (2025) traces the survival history: the underpricing should *widen* through calm, as κ_t erodes and the held views drift from the record, and *collapse* at disasters, when the cross-section is repriced. A physical-minus-priced tail gap that grows with time since the last disaster—alongside the residual-slope signature of Remark 1—is what the mechanism predicts and a fixed-distribution account does not; a gap that does not move once the current surface is controlled is the null that places the mechanism on the Markov side.

Appendix IA.2. Excess volatility

The price–dividend ratio $v(\Psi_t)$ moves because the cross-section moves; this subsection traces that movement into the second moment of returns. One structural fact organizes the result. The pricing state carries no Brownian part—in calm it moves by finite-variation selection drift and

the staggered calm-deadline switches, which are predictable and by Lemma 7 leave the traded price continuous, so the *price* jumps only at disasters (Proposition 3, Lemma 7)—hence $v(\Psi_t)$ is diffusion-free, and the belief contribution to return volatility is concentrated at disasters. It vanishes at a single non-revising belief and is generated by the lazy cross-section.

The tree is unlevered, so its dividend is the endowment and dividend growth is consumption growth; *excess* volatility is return volatility beyond this fundamental—the classic finding that prices move more than dividends justify (Shiller, 1981; LeRoy and Porter, 1981). We work with the return-variance form of that phenomenon: return volatility in excess of cash-flow (dividend-growth) volatility, the component the return/discount-rate decomposition of Campbell and Shiller (1988) attributes to discount-rate news—here, belief news. The variance difference carries not only the belief-repricing jump variance but also its covariance with the fundamental disaster jump; a disaster lowers cash flows, and where it also lowers valuations the interaction is positive and can materially amplify the excess (the term is retained exactly in $\mathcal{E}_t$ below, not dropped, so no sign is imposed). That belief heterogeneity in a complete market generates such excess volatility is the theme of Dumas et al. (2009); our contribution is to locate it—with σ constant and disagreement about the *disaster* law, the excess is concentrated in the jump component and is sustained by lazy revision rather than by sentiment or overconfidence. Write the instantaneous return variance as the rate of its predictable quadratic variation, $\text{Var}_t(dR_t^e)/dt = d\langle R^e \rangle_t/dt$.

The return and dividend variances share the diffusion σ^2 —with σ constant and v diffusion-free there is no excess *diffusive* volatility—and differ only in their disaster-jump terms, so the excess $\mathcal{E}_t$ is entirely a disaster object: a *signed* functional of the price-dividend jump $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$, hence of the full record-resolved state Ψ_t through v (Proposition IA.2.1). It is positive—excess volatility in the usual sense—exactly when the belief repricing amplifies the disaster return jump in mean square, and vanishes at a single non-revising belief, where $\psi_t \equiv 1$ and v is constant.

The return–dividend variance difference lives at disasters. At a single non-revising belief the price–dividend ratio is constant and the return inherits exactly the dividend’s variance; time variation in the disaster-belief cross-section makes v move and adds $\mathcal{E}_t$ —the Shiller (1981) and LeRoy and Porter (1981) excess when the repricing amplifies the jump, prices fluctuating beyond what fundamentals justify. Two features are special to the mechanism. First, σ is constant and v diffusion-free, so *none* of the excess is diffusive: it lives in the disaster component, the price reacting to a disaster by the belief-repricing factor ψ_t on top of the fundamental fall e^{-x} . This sits naturally with Shiller (1981)’s observation that information arriving in occasional lumps shows up as kurtosis rather than smooth variance—here the excess is a tail object by construction. Second, it is a pure belief term: it would vanish in a representative-agent disaster economy with the same physical (λ^*, F_J) , and it differs from continuous time-varying-disaster-risk models (Wachter, 2013), where a smoothly moving disaster probability injects diffusive return volatility; here the physical law is fixed and the repricing happens at disasters. It differs in mechanism, too, from the higher-order-expectations route to variance-bound violations (Kasa et al., 2014), where idiosyncratic signals make the map from shocks to prices non-invertible and agents must forecast the forecasts of others; here there are no private signals and no price–belief feedback—investors learn the disaster law from the common record, not others’ beliefs from prices—and the excess variance is the wealth-weighted power-mean repricing at a disaster. The sign and magnitude of $\mathcal{E}_t$ turn on ψ_t , hence on the operator v (Proposition 14) and the calibration (Section 5).

Fear, the equity premium, and excess volatility now share one driver: the moving cross-section S_t and the switch flow that revises it. At a disaster they move together—a jump in fear and premium (Section 4) and, when the repricing ψ_t amplifies the jump, a burst of excess return variance (Proposition IA.2.1)—after which the staggered calm revisions return the economy slowly toward complacency. Section C assembles these into the event-study analytics of a disaster and its aftermath.

A. Proof of excess volatility

Proposition IA.2.1 (Excess volatility is the belief-repricing jump): *With $P_t = C_t v(\Psi_t)$ and the cum-dividend return $dR_t^e = dP_t/P_{t-} + (C_t/P_t) dt$ (the price jump taken against the pre-jump price P_{t-} ; the finite-variation yield is unaffected by the jump, $P_t = P_{t-}$ within it):*

- (i) *Because v is diffusion-free, dR_t^e has diffusion σdW_t and, at a disaster of size x , jump $\Delta R_t^e(x) = e^{-x} \psi_t(x) - 1$ with $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$; hence*

$$\frac{\text{Var}_t(dR_t^e)}{dt} = \sigma^2 + \lambda^* \mathbb{E}_f[(e^{-J} \psi_t(J) - 1)^2], \quad \frac{\text{Var}_t(dC_t/C_t)}{dt} = \sigma^2 + \lambda^* \mathbb{E}_f[(e^{-J} - 1)^2], \quad (\text{IA.2.1})$$

the second being dividend (consumption) growth, with the same diffusion and disaster jump $e^{-x} - 1$.

- (ii) *The diffusive variances coincide: with σ constant and v diffusion-free there is no excess diffusive volatility. The excess is entirely a disaster term,*

$$\mathcal{E}_t := \frac{\text{Var}_t(dR_t^e) - \text{Var}_t(dC_t/C_t)}{dt} = \lambda^* \mathbb{E}_f[(e^{-J} \psi_t(J) - 1)^2 - (e^{-J} - 1)^2], \quad (\text{IA.2.2})$$

a functional of the price-dividend jump ψ_t alone, hence of the full record-resolved state Ψ_t through v (Propositions 14, 16). The term $\mathcal{E}_t$ is a signed difference of jump variances: $\mathcal{E}_t \geq 0$ exactly when the belief repricing amplifies the return jump in mean square, $\mathbb{E}_f[(e^{-J} \psi_t(J) - 1)^2] \geq \mathbb{E}_f[(e^{-J} - 1)^2]$, and is negative when the repricing damps it in mean square; we use “excess volatility” for the regime, dominant in the data, where the former holds.

- (iii) *At a constant cross-section—the representative-agent benchmark of a single non-revising belief (Remark 5)— v is constant, $\psi_t \equiv 1$, and $\mathcal{E}_t = 0$. Thus $\mathcal{E}_t$ vanishes at a single non-revising belief and is generated, in this model, by the time variation of the cross-section; we do not claim the converse that every moving cross-section yields $\mathcal{E}_t \neq 0$.*

Proof of Theorem 1. (i) By Proposition 2 and Proposition 16, $F_t(0) = \mathbb{E}[U_t \mid \mathcal{F}_t] = U_t = U^{\text{RA}} \kappa_t$ is a functional of S_t .

(ii) *The slope is the generator applied to U .* The fear factor $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t^\alpha]$ is a bounded functional of the cross-section ($g_t \in [\underline{r}^{1/\alpha}, \bar{r}^{1/\alpha}]$ on the compact $\mathcal{J} \times \Phi$), hence a bounded functional $U \circ \pi_\phi$ of the record-resolved state Ψ_t , which evolves as the piecewise-deterministic Markov process of Proposition 3—deterministic calm flow, disaster jumps at rate λ^* . By the extended-generator (Dynkin) identity for piecewise-deterministic Markov processes (Davis, 1984; Jacobsen, 2006), with $U \circ \pi_\phi$ in the domain (bounded, absolutely continuous along the calm flow),

$$\partial_s^+ F_t(0) = \frac{d}{ds} \mathbb{E}[U_{t+s} \mid \mathcal{F}_t] \Big|_{s=0+} = (\mathcal{A}_\Psi U)(\Psi_t) = \frac{dU_t}{dt} \Big|_{\text{calm}} + \lambda^* \int_{\mathcal{J}} (U(\Psi_t^x) - U(\Psi_{t-})) f(x) dx, \quad (\text{IA.2.3})$$

the generator splitting into the calm drift and the disaster compensator.

Calm drift. Differentiate $U_t = \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^\alpha f(x) dx$; the integrand's t -derivative $\alpha e^{\alpha x} g_t(x)^{\alpha-1} dg_t(x)$ is bounded uniformly on the compact $\mathcal{J}$ (as in Proposition 6), so dominated convergence justifies differentiating under the integral,

$$\frac{dU_t}{dt} \Big|_{\text{calm}} = \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t(x)^{\alpha-1} \frac{dg_t(x)}{dt} \Big|_{\text{calm}} f(x) dx. \quad (\text{IA.2.4})$$

The calm motion of $g_t(x) = \langle r(\cdot, x)^{1/\alpha}, S_t \rangle$ has two parts—the selection drift (18) and the calm switch flow (19) of Proposition 3, at $h = r(\cdot, x)^{1/\alpha}$:

$$\frac{dg_t(x)}{dt} \Big|_{\text{sel}} = -\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}), \quad \frac{dg_t(x)}{dt} \Big|_{\text{sw}} = \int_{\Phi} (r(\phi_t^*, x)^{1/\alpha} - r(\phi, x)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi), \quad (\text{IA.2.5})$$

the first by (E59), the second because a switcher relocating $\phi \rightarrow \phi_t^*$ changes her contribution to g from $r(\phi, x)^{1/\alpha}$ to $r(\phi_t^*, x)^{1/\alpha}$, at the wealth-weighted deadline rate Φ_t^{sw} . Substituting each

part of (IA.2.5) into (IA.2.4),

$$\begin{aligned}
\left. \frac{dU_t}{dt} \right|_{\text{sel}} &= \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t^{\alpha-1} \left(-\frac{1}{\alpha} \text{Cov}_{S_t}(\lambda, r(\cdot, x)^{1/\alpha}) \right) f dx = -\lambda^* \mathbb{E}_f[e^{\alpha J} g_t^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})] \\
&= -U_t b_{\text{sel}}(S_t), \\
\left. \frac{dU_t}{dt} \right|_{\text{sw}} &= \alpha \lambda^* \int_{\mathcal{J}} e^{\alpha x} g_t^{\alpha-1} \left(\int_{\Phi} (r(\phi_t^*, x)^{1/\alpha} - r(\phi, x)^{1/\alpha}) \Phi_t^{\text{sw}}(d\phi) \right) f dx \\
&= U_t c_{\text{sw}}(\Phi_t^{\text{sw}}),
\end{aligned} \tag{IA.2.6}$$

with $b_{\text{sel}}(S_t) = \mathbb{E}_f[e^{\alpha J} g_t^{\alpha-1} \text{Cov}_{S_t}(\lambda, r(\cdot, J)^{1/\alpha})] / \mathbb{E}_f[e^{\alpha J} g_t^{\alpha}] \geq 0$ and c_{sw} the ratio of the statement. *The factor α cancels in the selection term* ($\alpha \cdot \frac{1}{\alpha} = 1$, the share drift (18) carrying the $\frac{1}{\alpha}$), recovering the calm log-drift (25); *it does not cancel in the switch term*, where the deadline flux (19) carries no $\frac{1}{\alpha}$ —so c_{sw} retains the α , the asymmetry between the two calm channels. The switch term is signed $c_{\text{sw}} \leq 0$ by scheduled relief (Proposition 7: $r(\phi_t^*, \cdot) \leq r(\phi, \cdot)$ on the disaster-free switch set, $\Phi_t^{\text{sw}} \geq 0$, $\alpha > 0$).

Disaster compensator. The post-disaster state Ψ_t^x applies the reweight (24) and then relocates the triggered investors (D22), so the jump in U splits as

$$\lambda^* \int_{\mathcal{J}} (U(\Psi_t^x) - U(\Psi_{t-})) f dx = \underbrace{\lambda^* \Theta_{\text{rw}}(S_t)}_{\text{reweight}} + \underbrace{\lambda^* \Theta_{\text{sw}}(\Psi_t)}_{\text{relocation}}, \tag{IA.2.7}$$

$\Theta_{\text{rw}}(S_t)$ the change in $U = \lambda^* \mathbb{E}_f[e^{\alpha J} g^{\alpha}]$ produced by the size- x reweight of g_{t-} (24)—a functional of S_{t-} , since the reweight $r(\cdot, x)^{1/\alpha} / g_{t-}(x)$ and the reweighted aggregator are functionals of S_{t-} ; and $\Theta_{\text{sw}}(\Psi_t) \geq 0$ the additional change from relocating to ϕ_t^* those investors whose updated record leaves the typical set, at the record-dependent rate $\lambda^* \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D22)—a functional of Ψ_t , not of S_t , non-negative because the destination intensity is non-decreasing in the disaster count (Proposition 6(ii)).

Collecting. Substituting (IA.2.6)–(IA.2.7) into (IA.2.3),

$$\partial_s^+ F_t(0) = \underbrace{-U_t b_{\text{sel}}(S_t) + \lambda^* \Theta_{\text{rw}}(S_t)}_{\sigma(S_t)} + U_t c_{\text{sw}}(\Phi_t^{\text{sw}}) + \lambda^* \Theta_{\text{sw}}(\Psi_t), \quad (\text{IA.2.8})$$

the first group a functional of S_t through g_t (Proposition 16); the calm flow Φ_t^{sw} and the trigger propensity π are functionals of Ψ_t and not of S_t , since the deadline-hitting flow and the triggered set depend on the joint law of held models, ages, and records, which the held-model marginal does not fix. This is the decomposition of (ii).

(iii) For Ψ_t, Ψ'_t with $\pi_\phi \Psi_t = \pi_\phi \Psi'_t = S_t$: the levels coincide by (i), $F_t(0) = F'_t(0) = U_t$, and by (IA.2.8) the $\sigma(S_t)$ part of the slope coincides, leaving the two staggered-revision channels $U_t c_{\text{sw}} + \lambda^* \Theta_{\text{sw}}$. Conditioning on no disaster over the horizon suppresses Θ_{sw} and leaves the signed calm channel $U_t c_{\text{sw}} \leq 0$, so the state with the larger imminent deadline flow Φ_t^{sw} carries the weakly steeper decline, $\partial_s^+ F'_t(0) \leq \partial_s^+ F_t(0)$; unconditionally the two channels are oppositely signed. The wedge $F_t(s) - F'_t(s)$ is zero at $s = 0$ and first order in s ; its full-horizon law is Markov in Ψ_t , not autonomous in S_t (Proposition 14).

(iv) Under heterogeneous Bayesian learning each posterior is a sufficient statistic for its own future, so the wealth-weighted cross-section is Markov in itself; there is no first-passage deadline and no rejection trigger, hence neither Φ_t^{sw} nor Θ_{sw} , and (IA.2.8) reduces to $\partial_s^+ F_t(0) = \sigma(S_t)$, measurable in S_t . The forward curve is then pinned by the current cross-section at every horizon (Appendix I); an affine latent-state model and a representative filtering economy give the same conclusion through their own Markov states. $\square$

Proof of Proposition IA.2.1. Step 1 (return diffusion). With $P_t = C_t v(\Psi_t)$, take logs and differentiate:

$$\log P_t = \log C_t + \log v(\Psi_t), \quad d \log P_t = d \log C_t + d \log v(\Psi_t).$$

By Assumption 1 the only Brownian term in $d \log C_t$ is σdW_t . By Proposition 3 the record-resolved state Ψ_t has no Brownian part: in calm it moves by the finite-variation selection drift

(18) and the staggered revision flux—no positive Brownian mass revises at any single calm instant: by Lemma 6 the deadline $\Delta^*(\phi, \cdot)$ is strictly monotone in ε wherever the continuous size-deviance is informative, so with G atomless (Assumption 2) the induced deadline law is generically atomless and calm revisions stagger; on any flat level set of positive G -mass they synchronize, contributing a *predictable finite-variation* jump to Ψ_t , not a Brownian increment; by Lemma 7(iii) that predictable jump moves no traded price, so $v(\Psi_t)$ is continuous across it and the only *price* jumps—hence the only return jumps—are at disasters. Thus Ψ_t is a semimartingale with *no continuous local-martingale part*: its continuous component is the finite-variation selection drift, and all its martingale content is the compensated disaster jumps. Since v solves the integro-differential equation of Proposition 14, it is absolutely continuous along the deterministic calm flow—i.e. in the domain of the PDMP extended generator—so the Itô/Dynkin formula for piecewise-deterministic Markov processes (Davis, 1984; Jacobsen, 2006) represents $v(\Psi_t)$ as a semimartingale whose continuous part is the finite-variation generator drift and whose only martingale content is the compensated disaster jumps. In particular $v(\Psi_t)$ has no continuous local-martingale part— $\langle (v \circ \Psi)^c \rangle \equiv 0$ —and hence no Brownian part. So the Brownian part of $d \log P_t$ is σdW_t , and since $P = e^{\log P}$ shares the diffusion coefficient of $\log P$ (Itô), the diffusion of dP_t/P_t is σdW_t . The yield term $(C_t/P_t)dt$ is finite-variation, so $dR_t^c = dP_t/P_t + (C_t/P_t)dt$ has diffusion σdW_t .

Step 2 (return jump). At a disaster of size x , $C_t = C_{t-}e^{-x}$ (Assumption 1) and the record-resolved state moves $\Psi_{t-} \mapsto \Psi_t^x$, so $v(\Psi_t) = v(\Psi_{t-})\psi_t(x)$ with $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$. Then

$$\begin{aligned} \frac{P_t}{P_{t-}} &= \frac{C_t v(\Psi_t)}{C_{t-} v(\Psi_{t-})} = \frac{C_t}{C_{t-}} \cdot \frac{v(\Psi_t)}{v(\Psi_{t-})} \\ &= e^{-x} \psi_t(x), \end{aligned} \tag{IA.2.9}$$

line (IA.2.9) substituting $C_t/C_{t-} = e^{-x}$ and $v(\Psi_t)/v(\Psi_{t-}) = \psi_t(x)$; the yield term contributes

no jump, so

$$\Delta R_t^e(x) = \frac{P_t}{P_{t-}} - 1 = e^{-x}\psi_t(x) - 1,$$

matching ΔR_t^e in Proposition 15.

Step 3 (variances). Over $[t, t + dt]$, dR_t^e has the drift of Step 1–2, the diffusion σdW_t , and the compensated jump $\int_{\mathcal{J}} \Delta R_t^e(x)(\mu - \nu)(dt, dx)$. The Brownian and compensated-Poisson martingales are orthogonal, so their variances add. The Brownian part has variance $\sigma^2 dt$ ($d\langle W \rangle_t = dt$); the compensated-Poisson part $\int H(\mu - \nu)$ with $H = \Delta R_t^e$ has variance equal to its predictable variation $\int H^2 \nu = \lambda^* \int_{\mathcal{J}} \Delta R_t^e(x)^2 f(x) dx dt$ (valid for the bounded predictable H , $\mathcal{J}$ compact; Brémaud, 1981). Hence

$$\begin{aligned} \frac{\text{Var}_t(dR_t^e)}{dt} &= \sigma^2 + \lambda^* \int_{\mathcal{J}} (e^{-x}\psi_t(x) - 1)^2 f(x) dx \\ &= \sigma^2 + \lambda^* \mathbb{E}_f[(e^{-J}\psi_t(J) - 1)^2], \end{aligned} \tag{IA.2.10}$$

line (IA.2.10) writing the integral against f as $\mathbb{E}_f$. Consumption growth dC_t/C_t has the same diffusion σdW_t and disaster jump $C_t/C_{t-} - 1 = e^{-x} - 1$ (Assumption 1), so the same decomposition gives $\text{Var}_t(dC_t/C_t)/dt = \sigma^2 + \lambda^* \mathbb{E}_f[(e^{-J} - 1)^2]$. Together these are (IA.2.1), proving (i).

Step 4 (excess). Subtract the two lines of (IA.2.1); the common σ^2 cancels:

$$\begin{aligned} \mathcal{E}_t &= \lambda^* \mathbb{E}_f[(e^{-J}\psi_t(J) - 1)^2] - \lambda^* \mathbb{E}_f[(e^{-J} - 1)^2] \\ &= \lambda^* \mathbb{E}_f[(e^{-J}\psi_t(J) - 1)^2 - (e^{-J} - 1)^2], \end{aligned} \tag{IA.2.11}$$

line (IA.2.11) by linearity of $\mathbb{E}_f$; this is (IA.2.2), and it depends on Ψ_t only through ψ_t , hence through v (Proposition 14). This proves (ii).

Step 5 (benchmark). At the representative-agent benchmark of a single non-revising belief (Remark 5), Ψ_t may still update in its record and age coordinates, but the valuation is invari-

ant to those redundant coordinates here, so $v(\Psi_t)$ is constant and $\psi_t(x) = v(\Psi_{t-})/v(\Psi_t) = 1$ for every x , the integrand of (IA.2.11) is $(e^{-J} - 1)^2 - (e^{-J} - 1)^2 = 0$, and $\mathcal{E}_t = 0$. This proves (iii). $\square$

Appendix IA.3. A joint intensity–size illustration

Everything above runs in the intensity channel, where the held size law is the common truth and only the scalar multiplier $\kappa_t = G_t^\alpha$ moves; the risk-neutral disaster *shape* stays at the representative-agent benchmark. But Proposition 2 admits more. When investors disagree about disaster *size*, the risk-neutral law factors into the intensity multiplier κ_t —the headline bridge—and a mean-one *shape tilt* $R_t(x) = f_t^\mathbb{Q}(x)/f^{\mathbb{Q},\text{RA}}(x)$, and R_t departs from one exactly when the size cross-section is non-degenerate. The economically telling case is a cross-section whose wealth-weighted mean intensity is *correct* yet whose size beliefs disperse: the level κ_t is then near one, but R_t rises in x —the priced tail tilts toward larger disasters, in the direction the option smirk and the left-tail evidence display (Bollerslev and Todorov, 2011; Andersen et al., 2015). So the size channel is not a relabelling of the intensity channel; it moves the *shape* of the priced tail at fixed average intensity. We work this out in closed form for a 2×2 intensity–size menu—deriving the level–shape factorization, the monotone tilt, and a three-state numerical table—in Appendix A (Table I). The exercise demonstrates direction and mechanism, not a fit to a measured surface.

A. Details of the joint illustration

This appendix works out in closed form the size channel discussed in Section IA.3, at the Table I calibration.

A finite belief block. Take a 2×2 menu over intensity and size, $(\lambda, \theta) \in \{\lambda_{lo}, \lambda_{hi}\} \times \{\theta_0, \theta_H\}$, with the intensity menu of Table I, $\{\lambda_{lo}, \lambda_{hi}\} = \{0.01, 0.06\}$ around $\lambda^* = 0.03$. The size index

θ selects an exponential tilt of the physical size law, $f_\theta(x) \propto e^{\eta_\theta x} f(x)$, with $\eta_{\theta_0} = 0$ (the truth, $f_{\theta_0} = f$) and $\eta_{\theta_H} > 0$ (a fatter disaster tail); the held likelihood ratio is the closed form $r_\theta(x) = f_\theta(x)/f(x) = e^{\eta_\theta x}/\mathbb{E}_f[e^{\eta_\theta J}]$. With a finite menu the aggregator (12) is a finite sum,

$$g_t(x) = \sum_{\lambda \in \{\lambda_{lo}, \lambda_{hi}\}} \sum_{\theta \in \{\theta_0, \theta_H\}} s_{\lambda\theta, t} \left(\frac{\lambda}{\lambda^*} \right)^{1/\alpha} r_\theta(x)^{1/\alpha}, \quad (\text{IA.3.1})$$

and g_t is a finite sum of exponentials on the compact support $\mathcal{J}$; the pricing objects of Definition 2, built from $g_t(x)^\alpha$ and—for density-level objects—the physical f , are closed-form but not themselves finite exponential sums (the non-integer power α and the factor f preclude it).

Level and shape separate. Write the representative-agent risk-neutral size density as $f^{\mathbb{Q}, \text{RA}}(x) = e^{\alpha x} f(x)/\mathbb{E}_f[e^{\alpha J}]$, the value of (14) at $g_t \equiv 1$. The risk-neutral law then factors into a level and a shape,

$$U_t = U^{\text{RA}} \kappa_t, \quad f_t^{\mathbb{Q}}(x) = R_t(x) f^{\mathbb{Q}, \text{RA}}(x), \quad R_t(x) := \frac{f_t^{\mathbb{Q}}(x)}{f^{\mathbb{Q}, \text{RA}}(x)} = \frac{g_t(x)^\alpha}{\mathbb{E}_{f^{\mathbb{Q}, \text{RA}}}[g_t(J)^\alpha]}, \quad (\text{IA.3.2})$$

where the intensity multiplier $\kappa_t = \mathbb{E}_{f^{\mathbb{Q}, \text{RA}}}[g_t(J)^\alpha]$ is the bridge of Proposition 2 and the *shape tilt* R_t is a mean-one density ratio, $\mathbb{E}_{f^{\mathbb{Q}, \text{RA}}}[R_t(J)] = 1$. The two are independent summaries: κ_t carries the disaster *intensity* (the headline factorization), R_t the disaster-size *shape*. The intensity channel is exactly the corner $R_t \equiv 1$: the wealth-weighted size aggregator $g_t(x)$ is constant in x , so the shape stays at the representative-agent benchmark and only $\kappa_t = G_t^\alpha$ moves. The operative condition is on the *aggregate* tilt, not on dispersion as a primitive: homogeneous correct beliefs ($\theta_i \equiv \theta_0$) are sufficient for $R_t \equiv 1$, but a homogeneous *misspecified* size law ($\theta_i \equiv \theta \neq \theta_0$) makes R_t non-constant just as a non-degenerate size cross-section does. Whenever the aggregator changes shape relative to the correct-belief benchmark, $R_t \neq 1$ and the option-implied tail shape moves with it.

Monotone shape (the governing sign). With the exponential-tilt menu the aggregator is a

non-negative combination of two exponentials in x ,

$$g_t(x) = c_{0,t} + c_{H,t} e^{\eta_{\theta_H} x / \alpha}, \quad c_{0,t} = \sum_{\lambda} s_{\lambda \theta_{0,t}} \left(\frac{\lambda}{\lambda^*} \right)^{1/\alpha}, \quad c_{H,t} = \frac{\sum_{\lambda} s_{\lambda \theta_{H,t}} (\lambda / \lambda^*)^{1/\alpha}}{\mathbb{E}_f[e^{\eta_{\theta_H} J}]^{1/\alpha}}, \quad (\text{IA.3.3})$$

with $c_{0,t}, c_{H,t} \geq 0$ (the truth cells contribute $r_{\theta_0} \equiv 1$, the fat-tail cells the rising exponential). Differentiating, $g'_t(x) = c_{H,t}(\eta_{\theta_H}/\alpha) e^{\eta_{\theta_H} x / \alpha} \geq 0$ since $\eta_{\theta_H} > 0$, and κ_t does not depend on x , so $R'_t(x) = \alpha g_t(x)^{\alpha-1} g'_t(x) / \kappa_t$ carries the same sign: R_t is flat ($\equiv 1$) exactly when $c_{H,t} = 0$ (no fat-tail model held) and *strictly increasing* in x otherwise. The x -shape depends on the physical f only through the scalar normalisations $\mathbb{E}_f[e^{\eta_{\theta_H} J}]^{-1/\alpha}$ that set the relative weight $c_{H,t}$ of the exponential term; the *qualitative* monotonicity and the exponential-mixture family are f -independent, while the exact shape is not, and the size tilt η_{θ_H} with the held weights fixes the rest. The physical f enters (IA.3.3) and κ_t through these scalar normalisations; the tail-shape claims below are properties of (IA.3.3), not read off a fit.

Three states. Table I evaluates (IA.3.1)–(IA.3.2) at the Table I calibration ($\alpha = 3.5$, $\lambda^* = 0.03$; the physical size law pinned to its Table I moments $\mathbb{E}_f[e^{\alpha J}] = 2.82$ and mean 0.21 on $b \in [0.10, 0.55]$; size tilt $\eta_{\theta_H} = 3$). In the *intensity-only* state the held size law is the common truth, so $R_t \equiv 1$ and only the level is complacent ($\kappa_t < 1$). In the *mixed* state the wealth-weighted mean held intensity equals the truth, $\sum s_{\lambda} = \lambda^*$, yet R_t rises in x : even with the average disaster frequency right, size disagreement tilts the priced tail toward larger disasters, lifting the risk-neutral mean disaster size from the benchmark 0.29 to 0.34. In the *post-disaster* state both move— $\kappa_t > 1$ and R_t steeper—as wealth concentrates on the high-intensity, fat-tail models. The referee-relevant case is the middle one: the size channel is not a relabelling of the intensity channel, because it moves the tail shape at fixed average intensity.

Reading and scope. The statistic $R_t(x)$ is the operational content of the size half of Proposition 2: the risk-neutral disaster *intensity* factors as a scalar ($U_t = U^{\text{RA}} \kappa_t$), but the *size* law is a tilt, not a scalar rescaling; the statistic R_t isolates the *incremental* part of that tilt relative to the representative-agent risk-neutral benchmark ($R_t = f_t^{\mathbb{Q}} / f^{\mathbb{Q}, \text{RA}}$, mean-one under $f^{\mathbb{Q}, \text{RA}}$),

Cross-section	$\sum s\lambda$	level $\kappa_t = U_t/U^{\text{RA}}$	shape tilt $R_t(x)$	$\mathbb{E}_{f_t^{\mathbb{Q}}}[b]$	shape moves?
Intensity-only ($\theta \equiv \text{truth}$)	0.022	0.57	$\equiv 1$	0.29	no
Mixed ($\sum s\lambda = \lambda^*$)	0.030	0.91	increasing in x	0.34	yes
Post-disaster fear	0.052	2.26	increasing in x	0.37	yes

Table I: Joint intensity–size illustration at the Table I calibration ($\alpha = 3.5$, $\lambda^* = 0.03$, menu $\{0.01, 0.06\}$, size tilt $\eta_{\theta_H} = 3$; physical size law pinned to $\mathbb{E}_f[e^{\alpha J}] = 2.82$ and mean 0.21). The level κ_t is the intensity multiplier of the bridge (15); the shape tilt R_t (IA.3.2) is the risk-neutral size density relative to the representative-agent benchmark, with $\mathbb{E}_{f^{\mathbb{Q},\text{RA}}}[R_t] = 1$; $\mathbb{E}_{f_t^{\mathbb{Q}}}[b]$ is the risk-neutral mean disaster size (benchmark 0.29). The intensity-only row is the corner $R_t \equiv 1$ of the headline exercise; the mixed row moves the tail shape at the correct average intensity. Illustrative, not a calibration to option data.

while the total physical-to-risk-neutral size tilt is $f_t^{\mathbb{Q}}/f \propto e^{\alpha x} g_t(x)^\alpha$. Here $R_t \neq 1$ requires dispersed size beliefs ($R_t \equiv 1$ in the intensity channel; cf. Remark 4). An increasing R_t says the model does not merely scale the disaster tail—it changes *which* disaster sizes are priced most aggressively, fattening the deep tail after disasters in the direction the option smirk and the left-tail evidence of Bollerslev and Todorov (2011); Andersen et al. (2015) display. This is a closed-form demonstration of direction and mechanism, not a fit: the size menu and weights are illustrative instantaneous evaluations of the cross-section (not stationary laws, as in Section A), chosen to exhibit a non-degenerate R_t at the macro-disaster calibration, and the exercise claims no match to a measured option surface. As in the intensity channel, the physical law (λ^*, F_J) is fixed throughout; only the belief block moves.

Appendix IA.4. Lazy revision as state-dependent information rigidity

Read through the lens of the expectations-formation literature, the revision rule (Definition 1) is a *state-dependent sticky-information* mechanism: a held model is retained—and incoming data ignored—until a surprise triggers a re-selection. This appendix makes the mapping precise and places the model relative to the documented facts on forecast under- and over-reaction.

The conclusion is that lazy revision delivers *consensus underreaction* in the sense of Coibion and Gorodnichenko (2015), with a distinctive state-dependent rigidity, but *not* the individual overreaction of Bordalo et al. (2020); we are explicit about why.

Forecasts, the baseline, and the test

Fix a horizon h and take the forecast target to be the disaster count $Y_{t,h} = N_{t+h} - N_t$. Investor i , holding $\phi_i(t) = (\lambda_i(t), \theta_i(t))$, forecasts under her held model

$$F_t^i = \mathbb{E}^{\phi_i(t)}[Y_{t,h} \mid \mathcal{F}_t] = \lambda_i(t) h,$$

with forecast error and revision

$$e_{t,h}^i = Y_{t,h} - F_t^i = (N_{t+h} - N_t) - \lambda_i(t) h, \quad \text{FR}_t^i = F_t^i - F_{t-\Delta}^i = (\lambda_i(t) - \lambda_i(t - \Delta)) h,$$

the revision nonzero only across a switch (Definition 1, Step 2). The Coibion and Gorodnichenko (2015) test regresses the error on the revision,

$$e_{t,h} = a + b \text{FR}_t + u_t. \tag{IA.4.1}$$

Under full-information rational expectations the forecast is the true conditional expectation, the error is orthogonal to the information set and hence to the revision, and $b = 0$; $b > 0$ is *underreaction* (revisions do not go far enough, errors persist in their direction) and $b < 0$ *overreaction*.

The consensus underreacts

Write $m_t = \lambda(\text{MAP}(\rho_t))$ for the common re-selection target—the intensity of the full-history posterior mode every reviser adopts (Lemma 4)—and let the consensus held intensity be $\bar{\lambda}_t =$

$\int w_i \lambda_i(t) di$ for population weights w_i . Over $[t - \Delta, t]$ a fraction $\pi_t = h_t^{\text{rev}} \Delta + o(\Delta)$ of the population rejects and adopts m_t , at the revision intensity h_t^{rev} of Proposition 12, while the rest hold their models unchanged; to first order in Δ the consensus obeys the sticky-information recursion of Mankiw and Reis (2002),

$$\bar{\lambda}_t = (1 - \pi_t) \bar{\lambda}_{t-\Delta} + \pi_t m_t, \quad (\text{IA.4.2})$$

only the updating fraction π_t moving to the current target. This form takes the revisers to be representative of the population mean; revision here is state-dependent and selected, so in general the second term carries the *revisers'* pre-revision mean rather than the population $\bar{\lambda}_{t-\Delta}$, and (IA.4.2) holds exactly only when the two coincide. Otherwise the revisers' pre-revision mean differs from $\bar{\lambda}_{t-\Delta}$ by an $O(1)$ selection wedge; multiplied by $\pi_t = O(\Delta)$ this enters the drift at the same $O(\Delta)$ order as the consensus change itself, so the population-mean form is *not* a valid first-order approximation—the revisers' mean must be carried, and the reduction to $\bar{b} = (1 - \pi_t)/\pi_t$ below assumes the two means coincide (or differ by $o(1)$). With consensus forecast $\bar{F}_t = \bar{\lambda}_t h$ and target $m_t h$, the predictable part of the consensus error is the gap to the target. For that gap,

$$m_t h - \bar{F}_t = (m_t - \bar{\lambda}_t) h \quad (\text{IA.4.3})$$

$$= (m_t - (1 - \pi_t) \bar{\lambda}_{t-\Delta} - \pi_t m_t) h \quad (\text{IA.4.4})$$

$$= ((1 - \pi_t) m_t - (1 - \pi_t) \bar{\lambda}_{t-\Delta}) h \quad (\text{IA.4.5})$$

$$= (1 - \pi_t) (m_t - \bar{\lambda}_{t-\Delta}) h, \quad (\text{IA.4.6})$$

(IA.4.3) substituting $\bar{F}_t = \bar{\lambda}_t h$, (IA.4.4) substituting (IA.4.2) for $\bar{\lambda}_t$, (IA.4.5) collecting $m_t - \pi_t m_t = (1 - \pi_t)m_t$, and (IA.4.6) factoring $(1 - \pi_t)$. For the consensus revision,

$$\overline{\text{FR}}_t = (\bar{\lambda}_t - \bar{\lambda}_{t-\Delta})h \quad (\text{IA.4.7})$$

$$= ((1 - \pi_t)\bar{\lambda}_{t-\Delta} + \pi_t m_t - \bar{\lambda}_{t-\Delta})h \quad (\text{IA.4.8})$$

$$= (\pi_t m_t - \pi_t \bar{\lambda}_{t-\Delta})h \quad (\text{IA.4.9})$$

$$= \pi_t (m_t - \bar{\lambda}_{t-\Delta})h, \quad (\text{IA.4.10})$$

(IA.4.7) the definition of the consensus revision, (IA.4.8) substituting (IA.4.2), (IA.4.9) cancelling $(1 - \pi_t)\bar{\lambda}_{t-\Delta} - \bar{\lambda}_{t-\Delta} = -\pi_t \bar{\lambda}_{t-\Delta}$, and (IA.4.10) factoring π_t . When $m_t \neq \bar{\lambda}_{t-\Delta}$ the common factor $(m_t - \bar{\lambda}_{t-\Delta})h$ is nonzero, so dividing (IA.4.6) by (IA.4.10) cancels it:

$$\frac{m_t h - \bar{F}_t}{\overline{\text{FR}}_t} = \frac{1 - \pi_t}{\pi_t}, \quad \text{i.e.} \quad m_t h - \bar{F}_t = \frac{1 - \pi_t}{\pi_t} \overline{\text{FR}}_t. \quad (\text{IA.4.11})$$

Decompose the consensus error $\bar{e}_{t,h} = Y_{t,h} - \bar{F}_t$ into the target's error and this gap, then substitute (IA.4.11):

$$\bar{e}_{t,h} = (Y_{t,h} - m_t h) + (m_t h - \bar{F}_t) = (Y_{t,h} - m_t h) + \frac{1 - \pi_t}{\pi_t} \overline{\text{FR}}_t. \quad (\text{IA.4.12})$$

To the extent m_t is the agents' rational (full-information) target, the first term $Y_{t,h} - m_t h$ is orthogonal to the time- t revision $\overline{\text{FR}}_t$ (it is the target's own forecast error), so the population regression (IA.4.1) has slope equal to the coefficient on $\overline{\text{FR}}_t$ in (IA.4.12),

$$\bar{b} = \frac{1 - \pi_t}{\pi_t} > 0 \quad (0 < \pi_t < 1), \quad (\text{IA.4.13})$$

consensus underreaction; the sign is immediate from $0 < \pi_t < 1$ (with $\pi_t = h_t^{\text{rev}} \Delta + o(\Delta) \in (0, 1)$ for small Δ). The coefficient $(1 - \pi_t)/\pi_t$ is the *conditional*, state-dependent rigidity at t , not a single population slope: since π_t varies with the state, an unconditional Coibion–Gorodnichenko

regression estimates a covariance-weighted average of the conditional slopes $(1 - \pi_t)/\pi_t$ —the pooled OLS slope weights each state by the variance of the forecast revision there, not by π_t —and a target m_t that drifts with the same history that drives the revision contributes an omitted-variable term loading on the slope unless the regression conditions on the state. We therefore read $\bar{b} > 0$ as a conditional/local underreaction statement: any gap between m_t and the full-information forecast—mode versus mean, or the slow Berk drift of ρ_t toward ϕ^* (Berk, 1966)—enters the intercept only under that conditioning (orthogonality of the target’s forecast error to the revision); absent it, the gap can covary with the revision and shift the estimated slope.

State-dependent rigidity, and why this is not Bordalo–Gennaioli–Ma–Shleifer

Two features separate this from textbook sticky information and from the overreaction evidence. First, the updating fraction $\pi_t = h_t^{\text{rev}}\Delta$ is not a constant Calvo rate: the revision intensity h_t^{rev} combines the disaster-triggered rejection rate (Proposition 12, eq. (D22)), which spikes at disasters and is negligible in calm, with the cross-sectional calm-deadline flow—agents reaching their scheduled deadlines Δ^* —which runs at the low rate set by the deadline density. By (IA.4.13) the rigidity coefficient $\bar{b}_t = (1 - \pi_t)/\pi_t$ is therefore *countercyclical* relative to the fear/revision cycle (quiet spells versus disasters), not to aggregate activity: large in quiet times, when only the calm-deadline trickle revises, and compressing at disasters, when a mass of investors re-selects at once. Lazy revision predicts information rigidity that is highest precisely when the world has been calm—a testable refinement constant-rate sticky information does not deliver.

Second, this is underreaction, not the individual *overreaction* of Bordalo et al. (2020). The reason is structural. In Definition 1 the *test* reads the recency-weighted since-adoption window, but the re-selection *destination* is the mode of the *full-history* posterior ρ_t (9), which Berk-

converges to ϕ^* (Berk, 1966). A reviser catches up to the rational full-history belief; she does not overshoot it. Between revisions she is stale (individual underreaction); at a revision she corrects toward the rational mode. Individual overreaction in the sense of Bordalo et al. (2020)—a forecast that swings past the rational target in the direction of recent news—would require the destination itself to overweight the recent window, that is, a different re-selection rule than (9). The present model thus sits on the consensus-underreaction side of the evidence (Coibion and Gorodnichenko, 2015); reproducing the micro/macro discrepancy of Bordalo et al. (2020) is outside this rule, and we do not claim it.

Appendix IA.5. Further properties of the belief multiplier

This appendix collects two immediate consequences of the bridge (Proposition 2) and the power-mean kernel (Proposition 1), each derived from the primitives and scope-tagged. They formalize claims the main text uses in passing; none is invoked in the proof of any body result.

Corollary IA.5.1 (Pareto-weight invariance of the bridge): *Replace the symmetric allocation of Assumption 3(iv) by arbitrary admissible Pareto weights $\{\zeta_i\}$, $\zeta_i > 0$ with $\int_0^1 \zeta_i^{-1/\alpha} di \in (0, \infty)$. Then the equilibrium of Proposition 1 and the bridge of Proposition 2 hold verbatim,*

$$U_t = U^{\text{RA}} \kappa_t, \quad U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}], \quad \kappa_t = \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]},$$

with U^{RA} independent of $\{\zeta_i\}$. The weights enter only through the consumption shares $s_t^i = \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} / \Xi_t$ inside the aggregator g_t (Definition 2); the symmetric allocation fixes only which cross-section S_t —hence which multiplier κ_t —the kernel reads, not the form of the factorization.

Corollary IA.5.2 (Allocation-robustness of the wedge): *Fix any admissible Pareto weights $\{\zeta_i\}$*

as in Corollary [IA.5.1](#). Beyond the invariance of the bridge form, the wedge is disciplined independently of the allocation in three senses, and its stationary magnitude is independent of it altogether.

- (i) (The weights are endogenous, not chosen.) The aggregator weights s_t^i in g_t (Definition [2](#)) are the equilibrium consumption shares ([11](#)), pinned by realized belief performance (M_t^i) and the date-0 allocation; the weights $\{\zeta_i\}$ enter only as that initial condition, not as a free parameter in the pricing. The object the shares track is the wealth-weighted cross-section of held models.
- (ii) (Sign.) In the intensity channel (Corollary [1](#)), $\kappa_t = G_t^\alpha \leq \mathbb{E}_{S_t}[\lambda/\lambda^*]$ by Jensen for the power mean of order $1/\alpha < 1$, for any weights; hence the complacency $\kappa_t < 1$ at a cross-section correct on average ($\mathbb{E}_{S_t}[\lambda/\lambda^*] = 1$) holds for every allocation, strictly under dispersion.
- (iii) (Range.) For any weights κ_t lies in the menu-support interval $[\lambda_{\min}/\lambda^*, \lambda_{\max}/\lambda^*]$, with $\lambda_{\min}, \lambda_{\max}$ the extreme held intensities in the menu (a power mean lies between its extreme arguments); the wedge cannot exceed the spread of the held intensities, whatever the allocation.
- (iv) (Stationary magnitude.) Under turnover (Proposition [18](#)) the stationary law Ψ^∞ —hence the stationary law of κ_t and of every functional of Step 3—is independent of the date-0 allocation, which is encoded in the initial cross-section Ψ_0 and washes out (Proposition [18](#)(iii)); it is a functional of the newborn law Π_ϕ , the turnover rate Ω , and the newborn-endowment rule (Assumption [5](#)) alone. The stationary average wedge is therefore a property of the demographic primitives, not of the selected Negishi weights.

Proof of Corollary [IA.5.1](#). Proposition [1](#) is established for general ζ_i , the symmetric case $\zeta_i \equiv \zeta$ being a specialization; its state-price density $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha$ and shares $s_t^i = \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha} / \Xi_t$ carry the weights. By Definition [2](#) the aggregator $g_t(x) = \int_0^1 s_{t-}^i r_i(x; t)^{1/\alpha} di$ and the fear

factor $U_t = \lambda^* \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ are built from these shares, while the benchmark $U^{\text{RA}} = \lambda^* \mathbb{E}_f[e^{\alpha J}]$ is the value at $g_t \equiv 1$ (correct beliefs) and contains no ζ_i . Hence $U_t = U^{\text{RA}} \kappa_t$ with $\kappa_t = \mathbb{E}_f[e^{\alpha J} g_t^\alpha] / \mathbb{E}_f[e^{\alpha J}]$ for every admissible weight profile, and $\{\zeta_i\}$ affects only the shares inside g_t . $\square$

Proof of Corollary IA.5.2. (i) is (11). (ii) The map $t \mapsto t^{1/\alpha}$ is concave for $\alpha > 1$, so $G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^*)^{1/\alpha}] \leq (\mathbb{E}_{S_t}[\lambda/\lambda^*])^{1/\alpha}$ and $\kappa_t = G_t^\alpha \leq \mathbb{E}_{S_t}[\lambda/\lambda^*]$; at $\mathbb{E}_{S_t}[\lambda/\lambda^*] = 1$ this is $\kappa_t \leq 1$, strict unless S_t is degenerate (Jensen is strict for the strictly concave power under nondegenerate S_t). (iii) G_t is a weighted average of $(\lambda/\lambda^*)^{1/\alpha}$ over $\lambda \in [\lambda_{\min}, \lambda_{\max}]$, hence lies in $[(\lambda_{\min}/\lambda^*)^{1/\alpha}, (\lambda_{\max}/\lambda^*)^{1/\alpha}]$; raising to the α gives the stated interval. (iv) Proposition 18(iii) gives a unique Ψ^∞ independent of the initial cross-section Ψ_0 ; the date-0 weights $\{\zeta_i\}$ act only through Ψ_0 —they scale the date-0 shares (11)—so they do not enter Ψ^∞ , which depends on (Π_ϕ, Ω) and the birth-endowment rule through the functional $\mathfrak{F}$ of Proposition 18. $\square$

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